

Chapter 3 — Research Methodology

Hybrid Learning and Training Effectiveness at Union Bank of India

3.1 Objectives of the Study

The study examines how employees experience hybrid learning in a corporate training setting and how that experience is associated with later learning and work-related outcomes. The model developed in Chapter 2 treats Hybrid Learning and Trainer Competence as antecedents of Learner Engagement. Learner Engagement is then examined in relation to Training Effectiveness, Knowledge Retention, and Job Performance.

The primary objective of the study is:

To examine the relationships among Hybrid Learning, Trainer Competence, Learner Engagement, Training Effectiveness, Knowledge Retention, and Job Performance among employees of Union Bank of India.

The specific objectives are:

1. To assess employees' perceptions of Hybrid Learning, Trainer Competence, Learner Engagement, Training Effectiveness, Knowledge Retention, and Job Performance.
2. To examine the relationship of Hybrid Learning and Trainer Competence with Learner Engagement.
3. To examine the relationship between Learner Engagement and Training Effectiveness.
4. To examine the relationship between Training Effectiveness and Knowledge Retention.
5. To examine the relationship between Knowledge Retention and Job Performance.

Together, these objectives follow the sequence proposed in the conceptual model. The first objective provides a descriptive picture of employees' training experience. The remaining objectives examine whether employees who report a stronger hybrid-learning experience and more competent trainers also report greater engagement, more effective training, stronger retention, and better task-related performance.

3.2 Hypotheses of the Study

The study uses a path model based on relationships between the six constructs. Since the study uses a cross-sectional employee survey, the hypotheses are stated as relationships rather than as proof of causation. Each null hypothesis is tested at the 5 per cent level of significance against its corresponding alternative hypothesis.

No.	Null hypothesis (H ₀)	Alternative hypothesis
H1	There is no significant relationship between Hybrid Learning and Learner Engagement.	There is a significant positive relationship between Hybrid Learning and Learner Engagement.
H2	There is no significant relationship between Trainer Competence and Learner Engagement.	There is a significant positive relationship between Trainer Competence and Learner Engagement.
H3	There is no significant relationship between Learner Engagement and Training Effectiveness.	There is a significant positive relationship between Learner Engagement and Training Effectiveness.
H4	There is no significant relationship between Training Effectiveness and Knowledge Retention.	There is a significant positive relationship between Training Effectiveness and Knowledge Retention.

No.	Null hypothesis (H ₀)	Alternative hypothesis
H5	There is no significant relationship between Knowledge Retention and Job Performance.	There is a significant positive relationship between Knowledge Retention and Job Performance.

The first two hypotheses examine the conditions that may encourage employees to engage with hybrid training. The remaining three hypotheses examine the learning-to-performance sequence. H3 tests whether engagement is connected with effective learning, H4 tests whether effective learning is connected with retention, and H5 tests whether retained knowledge is connected with employees' task performance.

3.3 Definition of Variables

The study uses six constructs. Hybrid Learning and Trainer Competence are the independent variables. Learner Engagement is the immediate outcome of these training conditions. Training Effectiveness and Knowledge Retention are subsequent outcomes in the proposed path, while Job Performance is the final outcome.

Hybrid Learning refers to the planned use of digital and face-to-face learning as parts of the same training programme. It is measured through employees' views on flexibility, access to digital content, interaction, use of technology, and the integration of online and classroom sessions.

Trainer Competence refers to the trainer's ability to support learning. It includes subject knowledge, effective use of training technology, and the ability to encourage employee participation.

Learner Engagement refers to the employee's attention, effort, interest, and active participation during training. It is measured through attentiveness, involvement in activities, effort to understand the content, and participation in discussions or exercises.

Training Effectiveness refers to the extent to which employees consider the training relevant, clear, useful, and applicable to their work. It reflects immediate learning and perceived usefulness, rather than later retention or job performance.

Knowledge Retention refers to employees' ability to recall, explain, and retain important training content after the training session. It is measured through employees' own perceptions of their continuing recall of the learning.

Job Performance refers to employees' perceived task performance after training. The study focuses on accuracy, quality, efficiency, and the application of knowledge and skills in routine work.

The questionnaire contains 30 questions in total. Four questions confirm eligibility and describe the respondent group. The remaining 26 items measure the six variables: seven items for Hybrid Learning, three for Trainer Competence, four for Learner Engagement, and four each for Training Effectiveness, Knowledge Retention, and Job Performance. Employees respond to the construct items using a five-point agreement scale from Strongly Disagree to Strongly Agree. The questionnaire is included in the Appendix.

The questionnaire is inspired by the base paper of Uddin et al. (2026), which examines the pathway from blended learning to employee performance through learning-related development. Questions on Hybrid Learning, Training Effectiveness, Knowledge Retention, and Job Performance retain this workplace learning-to-performance logic. In the present study, Training Effectiveness represents employees' immediate learning of useful job-related content, while Knowledge Retention examines whether that learning remains available in later work situations. Trainer Competence and Learner Engagement are added to explain the training conditions that may support this pathway in the Union Bank of India setting.

3.4 Research Design

The study uses a quantitative, descriptive, and correlational research design. The descriptive part summarizes how employees perceive the six constructs. The correlational part examines whether the proposed relationships between the constructs are statistically significant.

The study is cross-sectional because responses are collected at one point in time. The unit of analysis is the individual employee. Each respondent reports their own experience of hybrid training, their engagement with training, and their perceptions of learning and work-related outcomes. The design identifies relationships between variables; it does not establish that one variable causes another.

This design is suitable because the study seeks to understand employees' views of an existing hybrid-training experience within one organizational setting. It allows the researcher to identify patterns in the data and to test whether the proposed relationships are supported among the selected respondents.

3.5 Sampling Methods

The population consists of Union Bank of India employees who have participated in training that combines digital and face-to-face learning. The study uses non-probability convenience sampling because access to a complete sampling frame is not available and participation depends on the availability and willingness of eligible employees.

To be included, an employee must have attended at least one training programme that included both a digital component and an in-person component. Employees who have experienced only one delivery mode are not included because they cannot assess the integration of the two modes.

The proposed sample size is approximately 80 respondents. This size is suitable for the planned descriptive, correlation, and regression analyses, including the first regression model, which examines Hybrid Learning and Trainer Competence together as predictors of Learner Engagement. The findings are interpreted as evidence from the selected respondents and are not treated as representative of every employee of the bank.

The sample is expected to include employees from different age groups, work-experience levels, and job levels, subject to their availability and consent. These background details are used to describe the sample; they are not treated as additional variables in the main conceptual model. The study remains focused on employees' experience of hybrid training rather than on differences between demographic groups.

3.6 Data Collection Procedures

Data collection is planned among eligible employees connected with Union Bank of India Regional Office Chennai South, located at 38 and 39 Whites Road, Express Estate, Royapettah, Chennai, Tamil Nadu 600014. The questionnaire is shared only after the required organizational and academic approval is obtained. Employees receive a brief explanation of the study, its academic purpose, and the voluntary nature of participation before they respond.

The study uses a structured, self-administered questionnaire. It may be distributed through an approved digital link and, where appropriate, through a printed copy. The questionnaire asks employees about their recent experience of hybrid training and does not require names, employee numbers, or any other direct personal identifier. Completed responses are used only for academic analysis. Participation is voluntary, and employees may decline to answer any question or withdraw before submitting the questionnaire.

The final dissertation records the actual data-collection period, number of questionnaires distributed, number returned, and number retained for analysis. Incomplete responses that do not contain sufficient information for construct-level analysis are excluded before statistical testing.

The questionnaire is completed by employees based on a training programme they have attended recently. This helps respondents provide answers from a specific and relevant experience. The researcher does not seek confidential customer information, performance records, or sensitive personal details. Responses are stored securely and are reported only in combined form so that an individual employee cannot be identified in the findings.

3.7 Research Questions

1. What are employees' perceptions of Hybrid Learning, Trainer Competence, Learner Engagement, Training Effectiveness, Knowledge Retention, and Job Performance?
2. Is Hybrid Learning associated with Learner Engagement?
3. Is Trainer Competence associated with Learner Engagement?
4. Is Learner Engagement associated with Training Effectiveness?
5. Is Training Effectiveness associated with Knowledge Retention?
6. Is Knowledge Retention associated with Job Performance?

3.8 Data Analysis Techniques

Responses are coded and checked for completeness before analysis. Descriptive statistics, including frequency distributions, means, and standard deviations, are used to describe the respondents and summarize the six constructs.

For each construct, item responses are combined to create a construct score after the reliability check. Higher scores indicate a more positive employee perception of that construct. This approach makes it possible to examine the proposed relationships at the construct level while retaining the individual questionnaire items for reliability assessment.

Cronbach's alpha is used to assess the internal consistency of the items within each construct. A value of .70 or above is treated as acceptable for the main analysis. Where a construct produces a lower value, the item wording and item-total statistics are reviewed before any decision is made about retaining or revising an item.

Pearson correlation is used to examine the direction and strength of the relationships between the construct scores. The hypotheses are tested through regression analysis. The first model uses multiple regression to examine whether Hybrid Learning and Trainer Competence are associated with Learner Engagement. Separate regression models then examine the relationships between Learner Engagement and Training Effectiveness, Training Effectiveness and Knowledge Retention, and Knowledge Retention and Job Performance.

For each hypothesis, the null hypothesis is rejected when the p-value is less than .05. When the p-value is .05 or greater, the null hypothesis is retained. The result is then explained in plain language using the exact variables tested. The analyses are reported with the relevant statistic, significance value, and a cautious interpretation of the relationship.

The analysis also considers the direction and practical size of the relationships. A statistically significant result is not described as proof that training alone caused a later outcome. The findings are interpreted alongside the limits of self-reported, cross-sectional data and the organizational context in which the survey is conducted.

3.9 Pilot Study, Reliability and Validity

Before the main survey, the questionnaire is piloted with approximately 20 eligible employees who are not included in the final sample. The pilot checks whether the questions are clear, relevant, and easy to answer. It also checks whether the response scale and questionnaire format can be completed without practical difficulty.

Reliability is assessed through Cronbach's alpha for each of the six constructs. The pilot results are reviewed construct by construct rather than relying only on one overall reliability value. If an item is unclear or does not fit its construct, its wording is reviewed before the main survey. Any change is recorded so that the final questionnaire and the analysis remain consistent.

The pilot also helps confirm whether the questionnaire length is reasonable for employees who are balancing training and routine work. It identifies any wording that may be unclear to employees from different job levels or with different levels of familiarity with digital learning. The same response scale is used across all construct items to make the questionnaire easier to complete and to support consistent analysis.

Content validity is supported by linking the questionnaire to the concepts reviewed in Chapters 1 and 2. The Hybrid Learning items reflect flexibility, access, interaction, technology use, and integration of digital and classroom learning. The Trainer Competence and Learner Engagement items reflect the trainer and engagement factors identified in workplace blended-learning research. The Training Effectiveness, Knowledge Retention, and Job Performance items reflect the sequence from immediate learning to retained knowledge and workplace application.

Face validity is assessed during the pilot by asking respondents whether the questions are understandable and relevant to their training experience. The research supervisor also reviews the questionnaire before the main survey. These steps help ensure that the instrument measures the constructs stated in the research objectives and hypotheses.

The study relies on employee perceptions and therefore does not replace objective measures such as examination scores, supervisor ratings, or bank performance records. This is considered when interpreting the results. The questionnaire is designed to provide a focused account of how employees experience hybrid learning and how they relate that experience to learning and performance in their work.